

Do Smart Money Concepts and ICT Signals Beat Chance? A Cluster-Robust Test on U.S. Equities, 2010–2026

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ABSTRACT

Smart Money Concepts (SMC) and ICT ("Inner Circle Trader") trading signals are widely circulated in retail trading communities but have not, to our knowledge, been subjected to a data-snooping-corrected out-of-sample-honest test in the academic or working-paper literature. We translate two publicly available Pine Script indicators implementing 42 SMC/ICT signal families line by line into Python and run them as event detectors across 501 S&P 500 constituents, 2010–2026, producing 4,326,101 events and 34,452,258 backtested (event $\times$ holding-period) observations across eight holding horizons. Because a purely long-biased signal shows a positive mean return from broad market drift alone over a long bull market, we benchmark every signal, combination, sector, and market-regime cut against a same-frequency, ticker-matched random-entry baseline run through identical backtest mechanics, rather than a zero-return null alone. Because trades on the same ticker share overlapping forward-return windows and are not independent draws, we use cluster-robust (by-ticker) standard errors rather than a naive per-trade test, and report Benjamini-Hochberg false-discovery-rate-corrected significance throughout. We find that we find no signal family, among 42 tested, with a statistically significant positive edge over the baseline after false-discovery-rate correction and cluster-robust inference. At the two-signal combination level, 0 of 60 tested pairs beat the baseline after correction. Results are further conditioned on sector and market regime, both of which materially change the picture. We interpret these results as evidence that, once tested against a fair frequency-matched benchmark with inference robust to the non-independence of overlapping trades, this signal family's publicly circulated appeal is not well supported by its own most direct translation into a testable form.

Keywords: technical analysis; Smart Money Concepts; ICT trading; data snooping; multiple hypothesis testing; cluster-robust inference; false discovery rate; backtest overfitting; random-entry baseline; market efficiency

JEL classification: G11, G12, G14, C12, C52, C58

1. Introduction

Smart Money Concepts (SMC) and ICT ("Inner Circle Trader") methodology have, over the past several years, become among the most widely circulated technical trading frameworks in retail trading communities, popularized primarily through video content and proprietary indicator scripts rather than peer-reviewed publication. Both frameworks describe price action in terms of institutional order flow -- order blocks, liquidity sweeps, fair value gaps, breaks of market structure -- and are typically presented and consumed without any out-of-sample statistical validation. This paper asks a narrow, testable question: translated faithfully from their most widely used public implementation into an auditable, reproducible research pipeline, and tested against S&P 500 daily data over a 16-year window, do these signal families show a forward-return edge that survives a fair comparison to chance?

"A fair comparison to chance" is doing real work in that sentence, and it is the paper's central methodological concern. A signal's mean forward return being significantly different from zero is, on its own, close to uninformative over a long bull market: almost any signal with a long bias will show a positive mean return purely from broad market drift, independent of whether it contains genuine predictive content. This is not a new observation -- it echoes a substantial literature on data-snooping bias in technical trading rule research beginning with Brock, Lakonishok, and LeBaron (1992) and formalized by White's (2000) Reality Check and Hansen's (2005) Superior Predictive Ability test -- but it is routinely absent from the backtests that circulate alongside retail technical-analysis content, including the source material this paper tests. We address it directly by constructing, for every ticker in our universe, a random-entry baseline run through mechanics identical to the signals under test, and by reporting significance against that baseline as the paper's primary result, with significance against a zero-return null reported only for context.

A second, related concern governs our statistical design. Our unit of observation -- a single (ticker, signal, holding-period) trade -- is not independent. A frequently firing signal generates trades with heavily overlapping forward-return windows on the same ticker, and market-wide events induce correlation across tickers on shared dates. Treating trades as independent draws, the conventional approach in retail backtesting software and in a meaningful share of the informal analysis this literature responds to, understates the true standard error of an estimated mean return and can manufacture statistical significance from noise as sample size grows. We address the within-ticker component of this problem with cluster-robust inference, clustering standard errors by ticker (Section 5.3), and disclose what remains uncorrected (Section 8).

The paper proceeds as follows. Section 2 situates this test within the technical-analysis data-snooping literature. Section 3 describes the data. Section 4 describes the source indicators and our translation methodology. Section 5 details backtest and statistical construction, including the baseline and cluster-robust testing framework. Section 6 reports results at the individual-signal, combination, sector, and market-regime level. Section 7 discusses what the results do and do not support. Section 8 states limitations plainly. Section 9 concludes.

2. Related literature

Brock, Lakonishok, and LeBaron (1992) is the modern starting point for this literature: testing 26 moving-average and trading-range-break rules on the Dow Jones Industrial Average from 1897 to 1986, they found excess returns inconsistent with several standard null models (random walk, AR(1), GARCH-M, exponential GARCH), providing influential early support for technical analysis. Sullivan, Timmermann, and White (1999) then applied White's (2000) Reality Check bootstrap -- explicitly designed to correct for the fact that a large universe of rules was searched before the best one was reported -- to an expanded universe of nearly 7,900 rules on a century of DJIA data. Their best in-sample rule remained significant even after correcting for data snooping across the full universe searched, but that edge vanished entirely in a subsequent out-of-sample decade: a result widely read as a caution that in-sample significance, even correctly adjusted for multiple testing, does not guarantee forward validity. Hansen's (2005) Superior Predictive Ability (SPA) test refined this framework with a studentized statistic and data-dependent null distribution, improving power and reducing sensitivity to the inclusion of obviously poor candidate rules.

Applying these corrected tests to markets beyond the DJIA has generally sharpened, rather than softened, the skeptical conclusion. Park and Irwin (2010) applied both the Reality Check and the SPA test to technical rules across 17 U.S. futures markets (1985-2004) and found statistically significant profits, after correcting for data snooping, in only two of the seventeen markets tested. Their companion survey (Park and Irwin 2007) concludes that while a majority of studies using these corrected methods do find some positive evidence, the evidence is frequently compromised by incomplete accounting for transaction costs and by search spaces that are not fully disclosed. Most directly relevant to the present paper, Bajgrowicz and Scaillet (2012) apply a false-discovery-rate framework -- the same class of multiple-testing correction used throughout this paper, following Benjamini and Hochberg (1995) -- to technical trading rules on the DJIA across more than a century of data, and find that while FDR control selects more genuinely outperforming rules than family-wise-error methods like the Reality Check, a persistence analysis shows investors could not have identified the future best-performing rules in advance, and realistic

transaction costs erase even the in-sample outperformance that survives correction. Outside of rule-based technical indicators specifically, Lo, Mamaysky, and Wang (2000) apply nonparametric pattern recognition to classical chart patterns on individual U.S. equities and find several patterns carry incremental statistical information over the unconditional return distribution -- a rare result offering qualified support for technical analysis at the individual-equity level, the same asset class tested here. At a broader remove, Harvey and Liu (2015) and Harvey, Liu, and Zhu (2016) argue that most claimed return anomalies in the empirical asset pricing literature, not only technical trading rules, are likely false positives once the number of models effectively tested is properly accounted for, and propose substantially higher significance thresholds as a result.

Three features distinguish the present study from this literature. First, SMC and ICT concepts -- despite wide practitioner circulation -- have not, to our knowledge, been subjected to a data-snooping-corrected test of this kind in the peer-reviewed or working-paper literature; the existing technical-analysis data-snooping literature tests classical indicators (moving averages, trading-range breaks, chart patterns), not this specific, more recent framework. Second, where the classical literature typically benchmarks against a zero-return or risk-adjusted-index null, we benchmark against a same-frequency random-entry baseline matched at the individual ticker level -- a stricter, more directly interpretable bar for a frequency-matched retail signal than an index-level null, and one that exposes the zero-null's weakness over a long bull market directly (Section 5). Third, we apply cluster-robust rather than naive standard errors given the overlapping-window structure of our trade-level data, a correction not uniformly present in the retail-facing backtesting practice this paper responds to.

3. Data

We use daily OHLCV data for the constituents of the S&P 500 index, sourced via the `yfinance` API, spanning 2010-01-01 through 2026-06-13 (approximately 16.4 years). The constituent list reflects current-day (2025/2026) index membership and weights applied retroactively across the full window; it is not a point-in-time-correct historical membership series (Section 8 discusses the resulting survivorship-bias limitation). Of 503 listed constituents, 501 returned usable price history; two (HONA, SATS) failed to download and were excluded. A single-ticker data-quality anomaly (HUBB, one day with an invalid OHLC relationship) was flagged and retained rather than silently patched. Full per-ticker coverage statistics are reported in the study's public data-quality export.

4. Signal translation and event construction

4.1 Source definitions

We translate two publicly available Pine Script indicators -- "Smart Money Concepts [LuxAlgo]" and "ICT Concepts [LuxAlgo]" -- line by line into Python, treating them as one combined methodology covering 42 distinct signal families spanning market-structure breaks (Break of Structure, Change of Character), order blocks, fair value gaps, liquidity sweeps, and related constructs. Every trigger condition was cross-referenced against each script's own alert-condition list to confirm complete coverage; two concepts named in general ICT literature (Rejection Blocks, Optimal Trade Entry) have no corresponding logic in either source script and are not implemented. Where the two source scripts define overlapping vocabulary under different exact logic (e.g., three distinct operational definitions of "Break of Structure"), each is retained as a separate, independently testable signal rather than merged, to avoid silently privileging one definition's threshold choices.

4.2 Detection and event construction

Each of the 42 signal detectors runs independently across every ticker in the usable universe, producing a master event table of (ticker, date, signal, direction) tuples. This produced 4,326,101 events. Detection logic operates exclusively in "historical" mode (Pine's 500-bar "Present" display window is not applicable to a full-history batch run), and several detectors that depend on strictly ordered historical swing points were validated against hand-constructed regression cases during development -- most notably a "stuck on the first unbroken swing" failure mode in the order-block detectors, where an old, never-broken swing point was incorrectly retained indefinitely rather than being superseded by the most recent confirmed swing, matching the source indicators' `swings()/leg()` semantics. A unit test reproducing this failure mode is maintained in the project's test suite.

5. Backtest construction and statistical methodology

5.1 Forward-return computation

For every detected event we compute forward returns at eight holding horizons -- 1, 2, 3, 5, 10, 20, 40, and 60 trading days -- using only the signal bar's own closing price for entry and strictly future bars for exit, maximum adverse excursion, and maximum favorable excursion. No information from the signal bar itself or earlier is used in any exit-side calculation. This produced 34,452,258

backtested (event x horizon) observations. Signals with no inherent long/short polarity in the source scripts (e.g., new-week and new-day opening-gap references, equal-highs/equal-lows reference levels) default to a long entry rather than being dropped; their reported performance should be read as "does price tend to rise after this reference level appears" rather than as a directional trading rule.

5.2 The random-entry baseline

A signal's mean forward return exceeding zero is a weak test of predictive content: over a 16-year window that includes one of the longest bull markets in the sample's history, a purely long-biased signal with no genuine information content will show a positive mean return from broad market drift alone. We therefore construct, for every ticker, a random-entry baseline: the same number of entries as a comparison signal (50 per ticker in our implementation), drawn uniformly at random from that ticker's trading calendar, and run through the identical backtest engine used for every SMC/ICT signal -- same forward-return windows, same holding horizons, same no-look-ahead construction. This produces a benchmark of "what does entering at this frequency, with no information content whatsoever, return on this specific stock over this specific period" -- the same-frequency, same-mechanics comparator that a signal must clear to be called informative, rather than merely long-biased.

Each ticker's random draw is seeded deterministically from a stable hash of the ticker symbol itself, ensuring reproducibility across runs while guaranteeing (to within a negligible collision probability given the size of the seed space relative to a 501-ticker universe) that no two tickers share an identical random draw.

5.3 Significance testing

For every signal family, combination, sector, and market-regime cell, we report two distinct hypothesis tests, deliberately kept separate throughout: a **zero-return null**, a one-sample test of whether the mean forward return differs from zero, and a **baseline comparison**, a two-sample test of whether the mean forward return differs from the ticker-matched random-entry baseline described above, at the same holding horizon.

Both tests use standard errors clustered by ticker rather than a naive per-trade estimator. This distinction matters here specifically because our unit of observation is not independent: a single ticker can contribute thousands of overlapping-window trades for a frequently firing signal (the most common signal in our sample fires over one million times across the universe), and those trades share substantial serial correlation through their overlapping forward-return windows.

Treating each trade as an independent draw -- the naive approach -- understates the true standard error of the mean and mechanically inflates apparent significance as sample size grows, independent of whether a true effect exists. We instead compute a cluster-robust ("sandwich") variance estimator for the sample mean, clustering by ticker, following the standard finite-sample-corrected formulation used for one-way-clustered inference in panel settings (Cameron, Gelbach, and Miller 2011; the same formula underlies Stata's default `vce(cluster)` for an intercept-only fit). This corrects for *within-ticker* serial correlation. It does not implement a full two-way (ticker-by-date) correction for *cross-ticker* correlation on shared market-wide dates (e.g., the March 2020 and 2022 drawdowns), which would tighten inference further in the same (conservative) direction; we treat this as a disclosed limitation rather than a completed correction (Section 8).

We report both a raw significance flag against each null (`significant_vs_zero`, `significant_vs_baseline` -- both two-sided) and a directional flag, `beats_baseline`, defined as significant *and* the excess return over the baseline positive. This distinction is not cosmetic: a signal can be significantly *different* from the baseline in either direction, and collapsing that into a single "significant" label without checking sign risks reporting signals that under-perform a random entry as if they were edges. Every p-value is corrected for multiple hypothesis testing within its natural family (all concepts jointly, all combinations jointly, and so on) via Benjamini-Hochberg false-discovery-rate control at $\alpha = 0.05$.

5.4 Combinations, sectors, and regimes

Concept combinations are constructed from same-direction signal pairs that co-occur on the same (ticker, date), subject to a minimum-occurrence floor (30 co-occurrences) and capped at the 60 most frequent pairs meeting that floor, to keep the multiple-testing correction meaningful rather than testing thousands of near-empty pairings. Sector groupings use a manually curated, static GICS-style mapping (Section 8 discusses this as a limitation). Market regimes are classified per-ticker from that ticker's own trailing 200-day simple moving average (trend) and realized-volatility-versus-own-history state (volatility), rather than an external market index, so that "bull market" and "high volatility" reflect each stock's own regime rather than a single shared macro state.

6. Results

6.1 Individual signal families

Of the 42 SMC/ICT signal families tested at the 10-day holding horizon, 42 show a mean return significantly different from zero after FDR correction — the weak test. Against the random-entry baseline, 35 of 42 concepts differ significantly, and every one of those 35 differs in the *negative* direction: none clears the baseline as a positive, significant edge. Table 1 is therefore empty by construction; Table 2 reports the signals significantly worse than a random entry at the same frequency.

No signal family beats the baseline at the current sample and FDR threshold; Table 1 is omitted.

Signal families significantly worse than the random-entry baseline, 10-day holding period.

Signal	N	Avg. return	Excess vs. baseline	p (FDR-adj.)
ict_fvg_bearish_filled	86,287	-1.42%	-2.26%	0.0000
ict_bos_bearish	42,780	-1.20%	-2.04%	0.0000
smc_fvg_bearish_formed	68,062	-1.09%	-1.93%	0.0000
smc_internal_bos_bearish	16,686	-1.03%	-1.87%	0.0000
ict_volume_imbalance_bearish	85,075	-0.94%	-1.77%	0.0000
ict_displacement_bearish	191,659	-0.90%	-1.74%	0.0000
ict_mss_bearish	36,776	-0.85%	-1.69%	0.0000
ict_fvg_bearish_formed	91,312	-0.88%	-1.72%	0.0000
ict_liquidity_sellside_swept	33,142	-0.88%	-1.72%	0.0000
ict_liquidity_sellside_pool_formed	52,038	-0.82%	-1.66%	0.0000

6.2 Two-signal combinations

At the two-signal combination level, 56 of 60 tested pairs (minimum 30 co-occurrences, top 60 by frequency) differ significantly from the baseline, and none does so in the positive direction; 56 are significantly worse.

No combination beats the baseline at the current sample and FDR threshold; Table 3 is omitted.

6.3 Sector breakdown

0 of 12 sectors show a positive point-estimate excess return over the random-entry baseline; 0 of those are statistically significant after cluster-robust, FDR-corrected testing (Table 4). Sector assignment uses a static, manually curated mapping (Section 8).

Sector-level average return versus the random-entry baseline, 10-day holding period.

Sector	Avg. return	Baseline avg. return	Excess vs. baseline	N tickers	Significant?
Consumer Staples	0.21%	0.42%	-0.20%	32	no
Unknown	0.36%	0.60%	-0.24%	48	no
Utilities	0.21%	0.50%	-0.29%	29	no
Financials	0.31%	0.64%	-0.33%	69	no
Materials	0.28%	0.64%	-0.37%	24	no
Real Estate	0.21%	0.63%	-0.43%	30	no
Industrials	0.37%	0.81%	-0.45%	65	no
Energy	0.33%	0.83%	-0.50%	20	no
Consumer Discretionary	0.40%	1.00%	-0.60%	49	no
Communication Services	0.28%	0.91%	-0.64%	20	no
Health Care	0.30%	1.05%	-0.75%	53	no
Information Technology	0.53%	1.52%	-0.99%	60	no

6.4 Market-regime breakdown

1 of 9 trend-by-volatility regime cells show a positive point-estimate excess return over baseline; 0 are statistically significant (Table 5). Regime classification uses each ticker's own trailing 200-day trend and realized-volatility-versus-own-history state, not an external market index.

Market-regime breakdown of average return versus the random-entry baseline.

Trend regime	Vol. regime	Avg. return	Baseline avg. return	Excess vs. baseline	N trades	Significant?
bear	high_vol	0.80%	2.26%	-1.46%	267,281	no
bear	low_vol	0.23%	0.14%	0.09%	15,569	no
bear	normal_vol	0.31%	0.87%	-0.56%	639,992	no
bull	high_vol	0.63%	1.39%	-0.76%	179,596	no
bull	low_vol	0.31%	0.48%	-0.17%	197,103	no
bull	normal_vol	0.29%	0.67%	-0.38%	2,094,334	no
sideways	high_vol	0.63%	1.95%	-1.32%	130,283	no
sideways	low_vol	0.25%	1.11%	-0.86%	27,139	no
sideways	normal_vol	0.20%	0.61%	-0.41%	763,386	no

7. Discussion

The central result of this paper is a clean null: once tested against a frequency-matched random-entry baseline with cluster-robust inference, none of the 42 SMC/ICT signal families examined here shows a statistically significant positive edge on S&P 500 equities over 2010–2026. This is a stronger conclusion than "the aggregate claim is not supported" — it is that not one of the specific, individually testable signal families that make up the aggregate survives the same test.

Two design choices in this paper are responsible for a materially more skeptical conclusion than a naive backtest of the same signals would produce, and both are worth stating plainly as the paper's main methodological contribution rather than only its results. First, benchmarking against a same-frequency random-entry baseline, rather than a zero-return null, matters enormously over a long bull market: every one of the 42 signal families tested here clears the zero-return null, which on its own would read as uniform support for the methodology. It is only against the baseline that the more sober picture in Section 6 emerges. Second, clustering standard errors by ticker — rather than treating millions of overlapping-window trades as independent draws — further tightens inference in the same direction. Both corrections point the same way: toward fewer, not more, apparently significant results. A retail backtest that skips either correction is, on the evidence here, likely to overstate the case for this signal family's predictive content.

Where the evidence is directionally more favorable — a positive point estimate that does not reach the paper's significance bar — the correct reading is not "probably works but underpowered." Section 6's per-sector and per-regime breakdowns show real heterogeneity in the sign of the excess return, which is consistent either with a genuine, narrow, conditional edge or with the ordinary variation expected across many correlated cuts of one dataset with no underlying effect. This paper's design cannot distinguish those two explanations with confidence, and does not attempt to; distinguishing them would require the walk-forward, held-out validation this paper does not implement (Section 8).

8. Limitations

- **Survivorship bias.** The universe reflects current-day S&P 500 membership and weights, applied retroactively across the full 2010-2026 window, not a point-in-time-correct historical constituent series. Constituents removed from the index during the sample period are entirely absent; constituents added more recently (e.g., several large technology and financial-services names) contribute their full available trading history despite not having been index members for most of that history. No point-in-time membership data was available to correct this.
- **Partial correction for cross-sectional correlation.** Standard errors are clustered by ticker, correcting for within-ticker serial correlation from overlapping holding windows. They are not two-way clustered by ticker and date, so correlation across tickers on shared market-wide dates is not separately corrected for. Implementing that correction would tighten inference further, not loosen it -- we view the results below as, if anything, a conservative upper bound on how many signals would survive a fully two-way-corrected test.
- **In-sample only.** All results are computed across the full sample window with no held-out, walk-forward validation split.
- **No transaction costs.** Reported returns do not account for commissions, slippage, market impact, or borrowing costs for short exposure; they are gross, idealized-execution returns.
- **Static sector mapping.** Sector groupings use a manually curated, best-effort mapping rather than a live point-in-time sector data source.
- **Undirected reference signals default long.** A small number of signals with no inherent directional polarity in the source indicators (new-week and new-day opening-gap levels, equal-highs/equal-lows) are scored as long entries by convention; their reported statistics describe price behavior following the reference level's appearance, not a validated directional trading rule.

- **Two literature concepts not implemented.** Rejection Blocks and Optimal Trade Entry, named in general ICT trading literature, have no corresponding logic in either source indicator and are therefore not tested.

9. Conclusion

We translate two of the most widely circulated Smart Money Concepts and ICT trading indicators into an auditable, reproducible Python research pipeline, and test all 42 resulting signal families against S&P 500 daily data, 2010–2026, using a same-frequency random-entry baseline and cluster-robust, false-discovery-rate-corrected significance testing throughout. The result does not support the aggregate, unconditional claim that these signals beat chance. No individual signal family survives the same test either. This paper's contribution is less the specific verdict on this one signal family, which is narrow and empirical by design, than the demonstration of how much that verdict depends on two choices — the null benchmark and the treatment of overlapping-trade correlation — that are frequently absent from the backtests this kind of trading methodology is popularized with. Code, data-processing pipeline, and full per-signal, per-combination, per-sector, and per-regime results are available in the accompanying public repository (Section 10).

10. Data and code availability

The complete data-processing pipeline, signal-detection code, backtest engine, statistical testing code, and full result tables (including every figure reported in this paper, at every holding horizon, not only the 10-day horizon reported in the tables above) are publicly available at <https://github.com/harshatangirala/smc-ict-research>. Price data is sourced via the public `yfinance` API and is not redistributed directly; the repository includes the ingestion pipeline needed to reproduce it, plus a lightweight cached data bundle sufficient to reproduce every table in this paper without re-downloading.

References

- Bajgrowicz, P., & Scaillet, O. (2012). Technical trading revisited: False discoveries, persistence tests, and transaction costs. *Journal of Financial Economics*, 106(3), 473-491.
- Benjamini, Y., & Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1), 289-300.

- Brock, W., Lakonishok, J., & LeBaron, B. (1992). Simple technical trading rules and the stochastic properties of stock returns. *The Journal of Finance*, 47(5), 1731-1764.
- Cameron, A. C., Gelbach, J. B., & Miller, D. L. (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics*, 29(2), 238-249.
- Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4), 365-380.
- Harvey, C. R., & Liu, Y. (2015). Backtesting. *The Journal of Portfolio Management*, 42(1), 13-28.
- Harvey, C. R., Liu, Y., & Zhu, H. (2016). ...and the cross-section of expected returns. *The Review of Financial Studies*, 29(1), 5-68.
- Lo, A. W., Mamaysky, H., & Wang, J. (2000). Foundations of technical analysis: Computational algorithms, statistical inference, and empirical implementation. *The Journal of Finance*, 55(4), 1705-1765.
- Park, C.-H., & Irwin, S. H. (2007). What do we know about the profitability of technical analysis? *Journal of Economic Surveys*, 21(4), 786-826.
- Park, C.-H., & Irwin, S. H. (2010). A reality check on technical trading rule profits in U.S. futures markets. *Journal of Futures Markets*, 30(7), 633-659.
- Sullivan, R., Timmermann, A., & White, H. (1999). Data-snooping, technical trading rule performance, and the bootstrap. *The Journal of Finance*, 54(5), 1647-1691.
- White, H. (2000). A reality check for data snooping. *Econometrica*, 68(5), 1097-1126.